



Configuration Parsing

The subject requires a NGINX-style configuration file that defines:

- Which ports to listen on
- Default error pages
- Maximum client body size
- Location-specific rules (methods, root, autoindex, upload, CGI)

Configuration file format (NGINX-inspired):

```
# start of server block
server {
# "listen" is a directive
    listen 8080;
    listen 8081;

    client_max_body_size 10M;

    error_page 404 /404.html;
    error_page 500 /50x.html;

    root /var/www/html;
    index index.html;

# start of location block
    location / {
        methods GET POST;
        autoindex off;
    }

    location /upload {
        methods POST;
        upload_enabled on;
    }
}
```

```
        upload_path /tmp/uploads;
    }

    location /cgi-bin {
        methods GET POST;
        cgi_extension .py;
        cgi_path /usr/bin/python3;
        root /usr/local/www/cgi-bin;
    }

    location /redirect {
        return 301 /new-location;
    }
}
```

Architecture of the parser:

```
Config File → Read File → Tokenizer → Parser → ServerConfig Struct
```

Common Parsing Pitfalls

- Missing semicolons → Parsing fails, throw error
- Unmatched braces → Track brace count, throw error
- Whitespace handling → Trim lines, handle tabs and spaces
- Comments → Skip lines starting with #
- Multiple locations same path → Later ones override (or error)
- Empty config file → Use defaults or error